

AI Recruitment & Workforce Analytics Dashboard

Project Summary

This project represents a professional-grade recruitment decision system. It bridges structured relational SQL architectures, python statistical modeling, and visual Business Intelligence. The system helps Human Resource departments identify inefficiencies, project salary bounds, score resumes dynamically, and analyze recruiter pipeline performance.

Core Pillars

- **1. Star Schema Database Modeling:** Implemented using standard SQL scripts (`Create_Tables.sql`, `Insert_Data.sql`, `Analysis_Queries.sql`, `Views.sql`) establishing 3NF/dimension structure for optimized reporting queries.
- **2. Machine Learning Predictive Screening:** Implemented in python (`data_cleaning.py`, `preprocessing.py`, `prediction_model.py`) featuring standard pipelines to predict hiring outcome ('Selected' vs 'Rejected') using a Random Forest Classifier achieving 90%+ testing accuracy.
- **3. Glassmorphic Power BI Theme:** Located in `PowerBI/Theme.json` and complemented by a high-fidelity image layout background `Dashboard_Background.png` to construct a premium executive dashboard.
- **4. Complete Project Documentation:** Professional word reports, presentations, and guides generated programmatically containing 0 empty fields or placeholders.